

Muhammad Bilal

Full-Stack Engineer — SaaS Architecture — AI Integration

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Summary

Full-stack engineer with 3 years of professional experience delivering production SaaS platforms and consumer-facing web applications end-to-end. Comfortable owning the full product lifecycle independently: system architecture, REST API design, database engineering, responsive frontends, CI/CD pipelines, DNS configuration, and PostgreSQL maintenance in live environments. Experienced in building AI-powered features including retrieval-augmented generation (RAG) pipelines, semantic search, and OpenAI integrations.

Skills

Languages:	JavaScript, TypeScript, Golang, Python, SQL, CSS
Frameworks/Libraries:	React, Next.js, Node.js, Express.js, Tailwind CSS, Material-UI
Databases:	PostgreSQL, MySQL, MongoDB, Firebase, pgvector
DevOps and Infrastructure:	Docker, GitHub Actions, CI/CD, Cloudflare, REST APIs, Cron Jobs
AI and Machine Learning:	RAG Architecture, pgvector, Mistral AI, OpenAI API, LLM Integration
Payments:	Stripe, Stripe Connect, Vendor Payouts, Order Reconciliation

Experience

Pixel-Square

Software Developer

June 2025 – Present

Germany

- Joined as an early engineer at a German software startup, taking end-to-end ownership across multiple client products, independently handling system architecture, full-stack development, deployment, and live production operations.
- Managed PostgreSQL database backups, cron job scheduling, and staging and production environment configuration across all active projects.
- Configured CI/CD pipelines via GitHub Actions for automated testing, linting, and repeatable deployments across environments.
- Managed DNS, SSL certificates, and subdomain routing via Cloudflare, ensuring zero-downtime availability for live clients.

Schul.software - School Management SaaS *(React, TypeScript, Node.js, PostgreSQL, Docker, REST API)*

- Architected and built a multi-tenant SaaS platform from scratch to production for German schools, implementing strict data isolation, per-school subdomain routing, and JWT-based authentication with role-based access control across admin, teacher, parent, and student portals.
- Designed a timetable engine with automated conflict resolution, teacher availability management, and substitute tracking, eliminating manual scheduling overhead for school administrators.
- Delivered end-to-end school operations features: grade tracking, attendance management, leave requests, announcements, and academic year management as a unified integrated platform.

Containerentsorgung - Multi-Vendor Marketplace *(Next.js, Node.js, TypeScript, PostgreSQL, Docker, Stripe)*

- Architected a multi-vendor marketplace with independent vendor storefronts, role-based access control, and per-vendor product, order, and fulfillment management.
- Implemented location-based service zone logic and dynamic shipping rule calculation based on customer geography.
- Integrated Stripe Connect for automated vendor payouts, platform commission splits, and end-to-end order reconciliation across multiple vendors.
- Deployed across staging and production environments using Docker and Cloudflare for consistent, reproducible releases.

Requirement Buddy - AI-Powered Ticket Generation *(Node.js, PostgreSQL, Mistral AI, RAG, pgvector)*

- Designed an end-to-end retrieval-augmented generation (RAG) pipeline using pgvector for semantic search, enabling development tickets to be auto-generated from natural language grounded in project history.
- Integrated Mistral AI embeddings to surface relevant historical context at query time, improving ticket accuracy and consistency across team workflows.
- Implemented role-based access control and secure authentication for multi-user, project-scoped team management.

Aurora Solutions

Software Engineer

July 2023 – April 2025

Sweden

- Led frontend engineering within an agile team building consumer-facing web platforms, contributing across the full stack on REST API integration, database queries, and end-to-end feature delivery from design to release.
- Participated in sprint planning, code reviews, and cross-functional technical discussions, translating product requirements into production-ready features across multiple release cycles.
- Improved application reliability and performance through database query optimization, caching strategies, and load testing.

Stakater Cloud MVP - Cloud Services Platform (*React, Golang, REST API, Docker*)

- Enhanced and maintained a cloud services web application, streamlining user sign-up, email verification, and automated cloud environment provisioning.
- Refined React frontend components for improved responsiveness and optimized onboarding workflows to strengthen security and user experience.
- Improved Golang backend performance for scalability, reliability, and efficient REST API communication under beta traffic.
- Monitored and resolved issues during beta testing, improving system stability ahead of full production release.

Cell Operative - Consumer Web Platform (*Next.js, Golang, Tailwind CSS, Docker, AWS, WebSockets*)

- Led frontend development across the platform, building accessible and responsive UI components in Next.js and Tailwind CSS with a focus on usability and cross-browser compatibility.
- Implemented user authentication and authorization via NextAuth.js and real-time bidirectional communication using WebSockets.
- Contributed to Golang backend services, collaborating on REST API design, data flow, and frontend-backend integration.
- Collaborated closely with backend engineers and product stakeholders across sprint cycles, contributing to technical decisions and maintaining consistent feature delivery.

ZUNTECH

React Developer Intern

Feb 2023 – Jun 2023

Pakistan

- Built and maintained responsive React.js UI components using Tailwind CSS as part of the frontend team, contributing to production-ready features across the development lifecycle.
- Collaborated with senior developers through code reviews, gaining hands-on experience with component architecture, state management, and agile workflows.
- Improved UI consistency and responsiveness across multiple views, supporting a better end-user experience on both desktop and mobile.

Al-Shifa Hospital - Healthcare Website (*React.js, Tailwind CSS*)

- Translated Figma design mockups for Al-Shifa Hospital into pixel-perfect, responsive React components using Tailwind CSS, delivering a professional healthcare web presence from designer specifications.
- Ensured cross-browser compatibility and mobile responsiveness across all implemented views.
- Collaborated with senior developers during code reviews, refining component structure and adherence to design system guidelines.

Education

University of Passau, Germany

M.Sc. in Computer Science

April 2025 – Present

FAST National University of Computer and Emerging Sciences

B.Sc. in Computer Science

Aug 2019 – Jun 2023

Languages

English (C1), German (A2 – Currently advancing to B1)